

HDOC – Day 29

C Programming Assignment

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Menu-Driven Array Operations in C

Problem Statement

Write a C program to perform the following menu-driven operations:

1. Count even numbers, odd numbers, positive numbers, negative numbers, and zero elements in an array.
2. Search for an element in an array using linear search.
3. Merge two arrays.
4. Exit.

Objective

To implement a menu-driven C program using arrays, loops, conditional statements, functions, and a switch statement for counting array elements, linear searching, and merging two arrays.

C Program

```
#include <stdio.h>
#define MAX 100
void countElements(void);
void linearSearch(void);
void mergeArrays(void);
int main(void)
{
    int choice;
    do {
        printf("\n===== ARRAY OPERATIONS MENU =====\n");
        printf("1. Count even, odd, positive, negative and zero elements\n");
        printf("2. Search an element using linear search\n");
```

```

printf("3. Merge two arrays\n");
printf("4. Exit\n");
printf("Enter your choice: ");
scanf("%d", &choice);
switch (choice) {
case 1:
countElements();
break;
case 2:
linearSearch();
break;
case 3:
mergeArrays();
break;
case 4:
printf("Exiting program...\n");
break;
default:
printf("Invalid choice. Please enter 1 to 4.\n");
}
} while (choice != 4);
return 0;
}

void countElements(void)
{
int a[MAX], n, i;
int even = 0, odd = 0, positive = 0, negative = 0, zero = 0;
printf("Enter number of elements: ");
scanf("%d", &n);
if (n <= 0 || n > MAX) {
printf("Invalid array size.\n");
return;
}
printf("Enter %d elements: ", n);
for (i = 0; i < n; i++)

```

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scanf("%d", &a[i]);
for (i = 0; i < n; i++) {
if (a[i] % 2 == 0)
even++;
else
odd++;
if (a[i] > 0)
positive++;
else if (a[i] < 0)
negative++;
else
zero++;
}
printf("Even elements : %d\n", even);
printf("Odd elements : %d\n", odd);
printf("Positive elements : %d\n", positive);
printf("Negative elements : %d\n", negative);
printf("Zero elements : %d\n", zero);
}
void linearSearch(void)
{
int a[MAX], n, key, i, position = -1;
printf("Enter number of elements: ");
scanf("%d", &n);
if (n <= 0 || n > MAX) {
printf("Invalid array size.\n");
return;
}
printf("Enter %d elements: ", n);
for (i = 0; i < n; i++)
scanf("%d", &a[i]);
printf("Enter element to search: ");
scanf("%d", &key);
for (i = 0; i < n; i++) {
if (a[i] == key) {

```

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position = i + 1;
break;
}
}
if (position != -1)
printf("Element %d found at position %d.\n", key, position);
else
printf("Element not found.\n");
}
void mergeArrays(void)
{
int a[MAX], b[MAX], c[2 * MAX];
int n1, n2, i, k = 0;
printf("Enter number of elements in first array: ");
scanf("%d", &n1);
if (n1 <= 0 || n1 > MAX) {
printf("Invalid array size.\n");
return;
}
printf("Enter %d elements of first array: ", n1);
for (i = 0; i < n1; i++)
scanf("%d", &a[i]);
printf("Enter number of elements in second array: ");
scanf("%d", &n2);
if (n2 <= 0 || n2 > MAX) {
printf("Invalid array size.\n");
return;
}
printf("Enter %d elements of second array: ", n2);
for (i = 0; i < n2; i++)
scanf("%d", &b[i]);
for (i = 0; i < n1; i++)
c[k++] = a[i];
for (i = 0; i < n2; i++)
c[k++] = b[i];

```

```
printf("Merged array: ");  
for (i = 0; i < k; i++)  
printf("%d ", c[i]);  
printf("\n");  
}
```

Sample Outputs

Test Case 1 - Counting

===== ARRAY OPERATIONS MENU =====

1. Count even, odd, positive, negative and zero elements
2. Search an element using linear search
3. Merge two arrays
4. Exit

Enter your choice: 1

Enter number of elements: 7

Enter 7 elements: 2 -3 0 4 -6 7 0

Even elements : 5

Odd elements : 2

Positive elements : 3

Negative elements : 2

Zero elements : 2

Test Case 2 - Linear Search (Found)

Enter your choice: 2

Enter number of elements: 5

Enter 5 elements: 10 20 30 40 50

Enter element to search: 30

Element 30 found at position 3.

Test Case 3 - Linear Search (Not Found)

Enter your choice: 2

Enter number of elements: 4

Enter 4 elements: 5 8 12 16

Enter element to search: 9

Element not found.

Test Case 4 - Merge Arrays

Enter your choice: 3

Enter number of elements in first array: 3

Enter 3 elements of first array: 1 2 3

Enter number of elements in second array: 4

Enter 4 elements of second array: 4 5 6 7

Merged array: 1 2 3 4 5 6 7

Test Case 5 - Exit

Enter your choice: 4

Exiting program...

Result

The menu-driven C program was implemented successfully. It correctly counts the required categories of array elements, searches an element using linear search, merges two arrays, and exits when the user selects option 4.